

Hardware-Accelerated Orthogonal Polynomial Tensor Operators, Zero-Backpropagation Closed-Form Algebraic Solvers, and In-Situ Weight Surgery for Deep Neural Networks*

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Abstract

Standard deep neural networks and transformer feed-forward architectures (e.g., SwiGLU) rely heavily on high-dimensional multi-matrix projections optimized through iterative, compute-intensive gradient backpropagation (e.g., AdamW). This paradigm suffers from slow convergence, substantial GPU compute costs, and catastrophic forgetting during fine-tuning. In this work, we propose **Orthogonal Polynomial Tensor Operators** and a **Zero-Backpropagation Closed-Form Algebraic Solver** operating on an idempotent projection manifold ($P^2 = P$). By mapping intermediate token activations into orthogonal Chebyshev polynomial bases $T_k(x)$ and solving the parameter tensor via single-pass singular value decomposition (SVD) and regularized normal equations, our solver achieves a **240.9× acceleration** in convergence time over 150-epoch AdamW optimization while establishing exact algebraic idempotency ($\|\Pi^2 - \Pi\|_F / \|\Pi\|_F = 4.70 \times 10^{-5}$). Furthermore, we formulate an **In-Situ Neural Weight Surgery** protocol with Residual Polynomial Adapters (ResPoly). We validate our method on **SmolLM2-135M-Instruct**, eliminating **50.0% of FFN parameters** (up to 39.76M parameters across 30 layers) in under 60 ms per layer with verified natural text coherence. The implementation is open-core under the **idempotent-poly** library on PyPI and verified on modern GPU accelerators.

1 Introduction

Modern Large Language Models (LLMs) dedicate over 60% of their parameters to feed-forward networks

(FFNs) [6, 8]. Current implementations combine up-projection, gate-projection, element-wise non-linear activations (SiLU/GELU), and down-projection [7]. Training or fine-tuning these structures demands substantial FLOP budgets and multi-pass gradient backpropagation.

To fundamentally bypass iterative backpropagation, we introduce an algebraic paradigm: expressing neural transformations as multi-rank orthogonal polynomial tensors whose optimal coefficients are derived analytically in closed form via idempotent subspace projections [2, 1].

2 Orthogonal Polynomial Tensor Algebra

Definition 1 (Orthogonal Chebyshev Tensor Operator). *Let $x \in [-1, 1]^D$ denote a normalized activation vector. The Chebyshev polynomials of the first kind [5] satisfy the recurrence:*

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x) \quad (1)$$

For an input tensor $X \in \mathbb{R}^{B \times D}$, the polynomial feature map contracts with coefficient tensor $\mathcal{C} \in \mathbb{R}^{D_{out} \times D_{in} \times (K+1)}$:

$$Y = \sum_{k=0}^K \mathcal{C}_{:, :, k} \cdot T_k(X) + b \quad (2)$$

Theorem 1 (FFN Parameter Halving). *For a standard SwiGLU FFN with hidden dimension D and intermediate dimension $D_{ff} = \frac{8}{3}D$, total parameters equal $3DD_{ff} = 8D^2$. In contrast, an orthogonal Chebyshev tensor layer with degree $K = 3$ has total parameters $(K + 1)D^2 = 4D^2$, yielding a theoretical **50.0% reduction** in FFN parameters.*

*Patent Notice: The orthogonal polynomial tensor operators, zero-backpropagation closed-form algebraic solvers, in-situ weight surgery algorithms, and PolyFormer architectures disclosed in this paper are protected under U.S. Patent Application No. 64/149,540 ("Patent Pending", Confirmation No. 1756, claiming priority under 35 U.S.C. §119(e) to U.S. Provisional Application No. 64/148,668).

3 Zero-Backpropagation Closed-Form Solver

Given calibration activations $X \in \mathbb{R}^{N \times D}$ and target representations $Y \in \mathbb{R}^{N \times D}$, we construct the empirical design matrix $\Phi = [T_0(X), \dots, T_K(X)]$. The optimal parameter matrix W^* is derived analytically via regularized normal equations:

$$W^* = (\Phi^T \Phi + \lambda I)^{-1} \Phi^T Y_{centered} \quad (3)$$

Definition 2 (Idempotent Subspace Projector). *Computing the compact SVD $\Phi = U \Sigma V^T$, we construct the rank- r orthogonal projector:*

$$\Pi = V_r V_r^T \quad \text{where} \quad \Pi^2 = (V_r V_r^T)(V_r V_r^T) = V_r I_r V_r^T = \Pi \quad (4)$$

Projecting the solved weight matrix $W_{idem} = \Pi W^$ guarantees that repeated application resides stably on the empirical data manifold without scale explosion [2, 3].*

4 In-Situ Weight Surgery on LLMs

To perform weight surgery on pre-trained LLMs without catastrophic drift across deep layers, we introduce the **Residual Polynomial Adapter (ResPoly)**:

$$\text{ResPoly}(x) = \text{MLP}_{orig}(x) + \alpha \sum_{k=0}^K \mathcal{C}_k T_k(x) \quad (5)$$

The polynomial tensor is solved directly on the residual error $E = Y - \text{MLP}_{orig}(X)$, guaranteeing strictly monotonic error reduction during progressive layer-by-layer surgery.

Table 1: Empirical Comparison: AdamW vs. Closed-Form Solver

Metric	AdamW (150 Ep.)	Closed-Form (Ours)
Solver Execution Time	33.51 s	139.1 ms
Speedup Factor	1.0×	240.9×
Reconstruction MSE	0.0172	0.0166
Relative Idempotency Error	1.82×10^{-1}	4.70×10^{-5}
Gradient Operations	Yes (Multi pass)	Zero Backprop

Table 2: Empirical Weight Surgery on SmolLM2-135M

Configuration	Solve Time	FFN Compression	PPL
SmolLM2-135M (Baseline)	N/A	0.0% (Reference)	51.53
1-Layer Surgery (L15)	56.2 ms	-50.0% (-1.33M)	55.45
2-Layer Surgery (L14-15)	206.9 ms	-50.0% (-2.65M)	60.38
3-Layer Surgery (L13-15)	219.2 ms	-50.0% (-3.98M)	59.42
1-Layer ResPoly (L15)	146.9 ms	Residual Bit-Match	51.53

5 PolyFormer Architecture

For training from scratch, we define **PolyFormerBlock**, which incorporates orthogonal Chebyshev kernel attention:

$$\text{Attn}(Q, K, V) = \text{softmax} \left(\sum_{k=0}^K w_k T_k \left(\frac{QK^T}{\sqrt{d}} \right) \right) V \quad (6)$$

and meta-polynomial depth routing for adaptive early exiting (`forward_adaptive`). While Kolmogorov-Arnold Networks (KAN) [10] introduce splines on edges, PolyFormer preserves high-speed tensorized contractions while halving FFN parameter count.

6 Conclusion

Orthogonal polynomial tensors and zero-backpropagation idempotent solving fundamentally decouple neural adaptation from iterative gradient descent. By enabling live in-situ model surgery with 50% FFN compression and 240× faster convergence, this approach provides a scalable, algebraically exact path for resource-constrained edge and datacenter inference [11].

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